

Nikolas “Dax” Manuel

Canadian Citizen

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EDUCATION

University of New Brunswick

Fredericton, NB

Bachelor of Science in Software Engineering | GPA: 3.7 (Dean's List).

Sep'24 – Apr '29

Relevant Coursework: Software Engineering, Programming in C, OOP in Java, Databases, Discrete Math, Calculus, Physics.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C, C++, HTML/CSS, SQL, Racket.

Frameworks/Libraries: Node.js, Express, React, Scikit-learn, Pandas, NumPy, OpenGL.

Tools: Git, GitHub, VS Code, MacOS, Windows, Linux, LLMs, Fusion 360, MS Office.

Courses/Certifications: [Code with Mosh: Ultimate JavaScript Series](#), [Code with Mosh: Data Structures and Algorithms](#).

EXPERIENCE

UNB Formula SAE

Sep 2025 - Present

Electrical Powertrain Member

- Building an electric racecar with a team of engineering students for FSAE.
- Working on electrical powertrain specifically low voltage systems to coordinate operational functions.

PROJECTS

2D Physics Engine | [GitHub](#)

C++, OpenGL

- Built a real-time 2D physics simulation in C++, applying object-oriented programming principles to model motion, forces, and collisions likewise to a realistic environment.
- Implemented rigid body dynamics for reusable objects such as gravity, restitution, and friction to simulate real objects.
- Visualized simulations at 60FPS using OpenGL to reach computer graphics standards.

Premier League Prediction Model | [GitHub](#)

Python, Scikit-Learn, Pandas, Numpy

- Achieved 60% accuracy on predicting Premier League soccer matches using linear regression through sci-kit learn.
- Preprocessed 700+ Premier League matches using pandas and numpy to optimize model performance.
- Tested the model on multiple props, such as final score, total goals, and both teams to score.

Photo Sharing App | [GitHub](#)

React-Native, AWS S3, Supabase, TypeScript

- Built a full stack cross-platform application for sharing photos within groups of friends using React-Native/Expo
- Implemented AWS S3 cloud storage to handle upload and access of user photos.
- Designed backend database architecture for user authentication and user data handling using Supabase.

CodeSimplify Chrome Extension | [GitHub](#) | [Demo Video](#)

NodeJS, Gemini API, HTML/CSS

- Built a full stack chrome extension that enables users to get code explanations when viewing code in GitHub.
- Designed strong LLM prompt structures to maximize context and clarity to optimize AI outputs with Gemini API.
- Designed a responsive and minimalistic popup user interface using HTML and CSS.
- Created RESTful API endpoints and rate limiting to manage user requests.

LEADERSHIP & ACTIVITIES

UNB Men's Varsity Soccer Player

Aug 2024 – Present

Youth Soccer Coach

April 2024 - Present

Fredericton District Soccer Association

Fredericton, NB

- Demonstrated strong leadership in managing groups of over 100 children in practices and summer camps.
- Provided mentorship and coaching to 500+ young athletes, to improve their skills understanding of the game
- Collaborated with other coaches, parents, and kids to ensure an inclusive, supportive, and positive team environment.

